

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1944

Commonwealth Chesapeake 345 kV
December 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1944. The Interconnection Customer proposes a 286.0 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on November 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 664 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$771,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Donner to Ninemile Pt 138 KV
Contingency: Cady to Dodd Rd 345 KV line
Loading: 112.1% (Rating: 477 MVA)
DFAX: 7.4%
Pre-project loading: 107.7%
2. Facility: Edgewater Station to C W Tippy 138 KV
Contingency: Bond-Elk Grove Florin 345/138kV Transformer
Loading: 111.1% (Rating: 354 MVA)
DFAX: 7.4%
Pre-project loading: 105.1%
3. Facility: Sutherland Springs to Hope Creek Substation 138 KV
Contingency: Bond-Elk Grove Florin 345 kV/138 kV Transformer Bank
Loading: 107.2% (Rating: 339 MVA)
DFAX: 7.4%
Pre-project loading: 101.0%
4. Facility: Bond-Elk Grove Florin Substation to Witmore Substation 138 KV
Contingency: Bond-Elk Grove Florin Substation 345 kV/138 kV Transformer Bank
Loading: 104.9% (Rating: 385 MVA)
DFAX: 7.4%
Pre-project loading: 99.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Donner to Ninemile Pt 138 KV	Rebuild line with higher-capacity conductor	\$2,277,000	5% (\$122,000)
Edgewater Station to C W Tippy 138 KV	Rebuild line with higher-capacity conductor	\$5,670,000	7% (\$379,000)
Sutherland Springs to Hope Crk 138 KV Line	Reconductor line	\$2,136,000	7% (\$143,000)
Bond-Elk Grove Florin Substation to Witmore Substation 138 KV	Reconductor line	\$1,899,000	7% (\$127,000)
	Total allocated to project		\$771,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.